

ActInf GuestStream 045.1 ~ Ramstead & Albarracin: "The inner screen model"

GuestStream | Ramstead & Albarracin | 1:53:02

hello and welcome it's June 19 2023
we're going to have a presentation and a
discussion with Maxwell ramstead and
Mala aburassen here so thank you Maxwell
and Mal for joining looking forward to
your presentation setting some of the
context and then thank you to all of our
guests you'll have an opportunity to
introduce yourself and give some
background when you first uh give
Reflections so
onto the slides Maxwell
excellent well maybe we now and I can
just briefly reintroduce ourselves for
uh those of us watching at home
um so hey folks uh my name is Maxwell
ramstead I am the uh director of
research at versus AI
um I am a scholar in the tradition of
active inference and an expert on the
free energy principle
um yeah and now would you please
sure so my name is I work at versus with
Maxwell and director director of product
for research and development but I'm
also doing a PhD in cognitive Computing
I worked in social sciences and in
artificial intelligence in the
artificial intelligence field and I've
specifically tried to apply active
inference to more socially grounded
questions and sometimes more abstract
and representational questions
diving all the way down to the
philosophy that can be attached to it

thank you Val

um so yeah I'm very pleased to present
this work uh today uh Mal and I will be
discussing our new pre-print

um called the inner screen model of

Consciousness and it is about it is

basically an application of the free

energy principle to the uh problem uh

the conundrum of Consciousness

um so uh just to give a little bit of

background and now I'll just be walking

through the slides if at any point you

want to uh jump in and you know add

complete correct uh please feel free to

do so

um so the uh this is work uh that

started in late last year

um and continued throughout the year

it's preoccupied us quite a bit actually

uh over the last while

um now and I are the lead authors on

this uh what has now become a series of

papers uh with uh one more pre-print

coming out hopefully before the end of

the months and another

in the works hope Pro hopefully before

the end of the year uh with the same

Core group of authors uh so obviously uh

Mao uh as well as Adam saffron Alex

Kiefer Brandon Klein Chris Fields uh and

Carl friston

um and uh the follow-up papers uh

include a bunch of other authors uh so

we've been uh working with some uh

pretty cool folks on this uh folks

listed here uh and others

um yeah and so the overall Lane across

the uh two or three papers uh is to

present something like a minimal
unifying model uh of Consciousness
premise on the free energy principle in
the paper that we're discussing today uh
we are presenting as I was uh intimating
a moment ago a model of Consciousness
that follows directly from applying the
free energy principle to well understood
human neurophysiology
um the take-home message of today's talk
is that uh well let me take one step
back there has been a lot of discussion
in the literature over the last decade
uh about whether the free energy
principle has anything uh unique to
teach us about Consciousness and there
have been responses Pro and Contra
um and I think that the question up
until recently was just unsettled uh we
would like to suggest a positive
response to that question
um and so as such one of the take-home
messages is that actually yes a model of
Consciousness can be directly derived
from the free energy principle uh
applied to explain
neuroanatomy and um I guess the the kind
of technical message the technical part
of the takeaway is that Consciousness
has or
um is isomorphic to a math a nested
holographic mathematical structure
um so uh this sounds like a mouthful
might sound a little crazy
um we are going to unpack this uh
presently
um I'll take some some now did you want
to add anything to that

no all good excellent

um so uh the original paper uh that we wrote turned into like a 24

000 word monstrosity of a paper uh and

it was uh elephant elephantine both in

the sense that it was very large

um and also in the sense that of the uh

Sufi parable of the Blind Men and the

Elephant what we're trying to do is in

some sense

um identify a parsimonious mathematical

structure that underwrites the different

accounts of Consciousness that have been

developed based on the free energy

principle

I'll spare you the original paper

structure

um I said a moment ago that we uh we're

developing a minimal unifying model this

is a notion that was proposed by our

friend and colleague Vania Visa uh uh I

believe a few years ago now in 2020

um and Vanya at the time was noticing

um that well the field of Consciousness

studies is replete with a bunch of

different models and theories

um and there seems to be no end in sight

to the multiplication of these theories

and uh you know the idea is maybe what

the field needs is not one additional

you know n plus one Theory Of

Consciousness but rather a minimal

unifying model uh by which Vanya means a

model that specifies only necessary

properties of Consciousness so this

doesn't come with like a strong

sufficiency claim really like what are

the kind of rudimentary building blocks

that are necessary for Consciousness in general

um this model would have determinable descriptions that can be made more specific so it's kind of a base model that can be expanded to cover different varieties of conscious experience and finally that you know it's minimal and unifying in the sense that it integrates a current approaches to Consciousness in no small part by highlighting their common assumptions

um so you know kind of pointing to a least common denominator of all existing accounts and so what we have tried to do in this paper is to uh well in this series of papers is to do uh engage in this analysis and this uh mummy analysis if you will for theories of

Consciousness uh premised on the free energy principle directly so it's worth quickly discussing the free energy principle uh I'm almost embarrassed to do this at the Active inference Institute uh but it's always worth going through this very quickly

um so there's a standard distinction in physics that one can leverage between Dynamics mechanics and principles the free energy Principle as it says on the tin is a principle

um so principles are kind of the foundation of a hierarchy of theory building in science at the as it were top of the hierarchy uh our formal descriptions of behavior what are known as Dynamics so if you're applying dynamical systems theory to understand

uh you know the time evolution of some system you're in the realm of Dynamics uh arguably folks like Kepler and Galileo were also in the realm of Dynamics so I see some empirical phenomenon and I produce a formal description of that behavior mechanics comes into play when we move from Kepler and Galileo to Newton when we move from merely describing some Behavior to providing equations of motion that allow us to explain why that behavior has the particular shape that it does um so uh you know classical Mechanics for example gives us an account of gravitational force and explains why orbits are shaped the way they do so if Galileo and Kepler just noticed the shape of orbits and provided a formal description of them Newton explains where that shape comes from in some sense and principles in turn explain where mechanics come from so classical mechanics come from the principle of least action or stationary action which uh very roughly speaking is the principle According to which uh and no more or no less energy is used than necessary to perform some physical movements that kinetic and potential energies balance out such that the true Paths of the system through its state space are those for which that balance equals zero the principle of least action uh explains where uh classical mechanics comes from in some sense and similarly uh the free energy principle

is at the basis of another kind of mechanics uh that is becoming known as Bayesian mechanics you can think of the 300g principle as having the same relation to Bayesian mechanics as the principle of least action has the classical mechanics and Bayesian mechanics is the physics of probabilistic beliefs so um it's a physics that connects the uh properties of the system that you're considering the physical properties of the system so for in particular the information entropy of uh states of the system to the uh thermodynamics sorry to the in it connects the thermodynamic entropy of the states of a system to the information entropy of the beliefs that a system has about the the uh systems to which it is coupled here A probabilistic belief uh means a probability density uh uh in a very general sense we're not assuming some kind of contentful notion of belief like you might find in uh philosophy really this is about kind of a tracking relation but we'll get that to that in uh actually just a few slides there are two main formulations of the free energy principle in the literature the classical and the quantum both rest on this apparatus of the Markov blanket uh the Markov blanket definitionally is uh the degrees of freedom that separate but couple uh two systems or rather two things within a larger system it's effectively a statistical boundary uh that allows us to say that well given

this boundary the inside is independent
of the outside

um the way that we uh Express the
dependencies so taking one step back
quickly this is a dynamic systems
approach to uh to cognition to a
self-organizing systems and to physical
systems more generally uh it is
dynamical systems approach in the sense
that we're you know using the tools of
dynamical systems theory to examine the
time evolution of a system it's
trajectory in its state space and in
this context a generative model is used
to encode the relations of dependents
that obtain in the time evolution of the
system so this is important and I've
been uh you know uh the kind of making
this point uh like uh maybe Ad nauseam
for the last I guess half decade but the
generative model is not a model that you
have in your head

it is a uh it is a statistical model
that represents the conditional
dependence structure of the entire
system that you're considering so this
is an an environment agent system uh in
the broadest sense

um and uh the free energy principle says
that if a specific sparseness structure
obtains uh so if things are disconnected
in a special way uh then uh it'll look
as if uh the subsets that are
disconnected are tracking each other
um so uh more technically uh we
partition our system into particles uh
particles are basically internal States
shrouded behind their Markov blanket

where the Markov blanket is composed of sensory States and active States sensory states are affected by internal States but are not affected by the internal States and active States affect external state but are not affected by external States uh this uh kind of this separation of causal dependence uh and this structure uh intervening between the two bulk uh subsets of the system is really critical

um the free energy principle says if my generative model which encodes all of the dependencies of the system if my generative model contains a Markov blanket in the sense that we just defined then we uh are basically licensed in our interpretation of internal States as basically tracking external states where tracking has a specific mathematical definition in this context tracking means to encode the parameters of some probabilistic belief about external States so to the extent that we want to call this representation we should note that this is not a representation in the typical teliosemantic sense of having some internal States track some external features

um it what we're what we're saying is that basically if I have a Markov blanket then my internal states are basically shaping a belief about the external world and I think that's really critical to understand um and this is where the inference and

the modeling that comes from this kind of tracking relationship formally speaking is approximately inference so basically in a nutshell the free energy principle says in our physical universe as we understand it with the the usual mechanics running in the background right classical relativistic Quantum and statistical uh if boundaries exist in the physical system then the bulk across the boundary is going to be tracking whatever is beyond the boundary so in a nutshell this is what the free energy principle says now I noticed you put some stuff in the chat would you mind using your voice to complement that yeah I was just saying that this part that you were just explaining often seems kind of as etheric to people um but it's critical to understand the sparseness of the coupling because then you kind of start understanding where perspective comes from and if you have perspective you also have uh the the capacity to um to assess causality um starting from a point and extending outwards to something else and this is true from any so that's that's what I wanted to add thank you Mama I think that's useful addition um yeah so this is the classical formulation of the free energy principle um there is a new hit funky uh uh formulation that has become available over the last half decade due to uh Chris Fields uh Jim glazebrook and their

colleagues these so-called Quantum information theoretic formulation of the fear not this doesn't appeal directly to Quantum Mechanics rather it appeals to uh the tools that have been used to extend information theory in the context of quantum mechanics which allow you to calculate like all sorts of funky wave equations and propagation operators and all that we don't have to get into the details basically it's just a way to reframe what I just said so what I just said is if there's a boundary in my physical system then the things across the boundary will look as if they're tracking each other another way of saying that is um well this boundary is a kind of screen uh green uh takes on a particular sense in Quantum Quantum information uh Theory uh we don't necessarily have to get into the details of it but basically um you can think of these degrees of freedom that couple the two systems as a kind of probabilistic surface and you can think of the bulk across either side of the boundary as in alternation reading and writing information onto the blanket so uh any Markov blanket whatsoever can be construed as a classical information Channel or a screen and in effect uh all of the classical information that you need to describe the way that things across the boundary so the so-called bulk of the system couple to uh one another is contained on the boundary it encodes the classical information that's

necessary to describe the couplings uh
within the system

um so uh we this is the this is
holography not in the sense of um this
is the Hologram not in the sense of
holography although we might discuss
senses in which these things are
intertwined

um we're we're using the word hologram
in the sense of the holographic
principle uh from physics the
holographic principle is a principle
that was originally uh discovered in the
context of black hole thermodynamics and
what it basically tells us is

necessarily from the point of view of an
external Observer some bulk in in
physical space can only contain as much
information as can fit onto its surface
uh so this Echoes what we were just
discussing about Markov blankets the
physical reason is that if a bulb
collapses into a black hole

um and the bulk contained any more
information than could be fit onto its
boundary uh then we would basically be
losing information and that would
violate uh the conservation of classical
information also known as the principle
of unitarity in quantum mechanics which
basically says that information is never
lost

um so uh we're using the word hologram
in that sense this is different from the
sense of hologram uh I mean everyone has
seen like the Hologram of Tupac or does
that make me look seem old at the Super
Bowl a few years ago or maybe Pokemon

cards maybe that also makes me seem old
now I don't know what the kids are doing
but you've seen like red baseball cards
and stuff like that with different
superimposed like picture layers that
become visible as you move that's not
the sense of hologram that's at stake
although maybe so we'll get to why but
let's we'll focus on the Hologram
theoretic formulation
so to summarize in the Hologram
theoretic hologram holographic principle
version of the free energy principle the
quantum information theoretic version of
the principle all of the classical
information that I need as an external
Observer to characterize the coupling
between two subsystems of a larger
system exist on the boundary
um and that's a pretty remarkable
physical fact
and a few years ago uh Chris Fields Jim
glazebrook and Mike Levin proposed that
maybe uh this kind of uh information
concentrating bottleneck architecture uh
also has application to uh Consciousness
in the form of an internal screen so the
idea was that you know the the
information that I bring to bear to
parse my sensory stream and make sense
of what is constantly bombarding me it
has to live somewhere
and so uh the idea is well it lives in
this inner screen so you basically have
an external Markov blanket which is a
Markov blanket just in the usual sense
and then
um the idea is well within the set of

internal States

um that uh are kind of partitioned off
from the organisms due to the organisms
Markov blanket then you would have an
internal Markov blanket structure within
those internal States

um this is prediction six from the great
paper minimal physicalism internal
awareness requires internal boundaries
and there's this interesting connection
to integrated information Theory I'm I I
am agnostic with respect to IIT so we're
not going to go there today but for
those of you who are interested this
kind of inner loop mechanism seems to
generate positive integrated information
so there's an interesting connection
there

um before I move on to neurophysiology
I'll hand things over to Mao if you you
wanted to add anything yeah actually it
was literally about to type something um
that this is actually also connected to
some Neo materialist formulations of
um folding onto onto itself so maybe
this is something worth digging into a
little later in the talk

awesome uh then we shall and I I believe
that Ali is also a co-author uh on the
paper and we have him so uh
Congratulations by the way on the
pre-print I I saw it released and Mao
has told me about it I'm really happy to
see uh such serious uh philosophical
work being done with the free energy
Principle as the basis I think you
should be very happy with the end result
and proud of but uh now uh let me get

back to uh my monologue

um so our uh rather uh our now and my uh
diet it's not really a dialogue it's
sort of like a I don't know how to call
it anyway

um

I'm thinking in French I have the words
in French I promise

um so

um now that we've you know reviewed in a
very much too short way to brutally fast
fashion uh the free energy principle and
its core formulations uh we're now going
to derive a model of Consciousness by
applying it to the known neurophysiology
of the brain are you ready

you buckled up okay good

um so uh we have just said that the fep
can be used to model things that have
boundaries I submit to you that the
brain is a thing with boundaries and
importantly a lot many several internal
boundaries

um so in particular we know that the
brain has a very sparse structure

um so you know if you think of uh just
the raw numbers right there are roughly
on the order of

150 160 billion neurons in the brain

um but each neuron only makes about uh
you know something like a five I want to
say five figures because I've spent uh
so uh 10 10 to the uh five uh
connections right so like roughly like I
think that

my you know my neurophysiology is a bit
far behind me but it's something on the
order of like a few thousand to maybe

tens of thousands of connections per neuron so obviously if you just think about it most neurons are not connected to each other not directly and furthermore the brain has a hierarchical structure and when you think about it what is a hierarchy a hierarchy is a sparse connectivity structure right so if you think about a hierarchy of connectivity what having a hierarchical structure means at the end of the day this isn't some statement about power dynamics where things at the top of the hierarchy are more important than things at the bottom your cortical neurons don't drain more juice uh than you know other uh parts of the brain the neurons in those parts uh no what we're talking about is a regular sparse pattern of connectivity where I only make connections to the layers directly below me and to the layers directly above me what I want to suggest now is that you can understand the sparseness as a set of nested Markle blankets and that's going to do a lot of heavy lifting for us um so the discussions here are based on two papers uh one by uh fields and colleagues called neurons as hierarchies of quantum reference frames it's really cool I recommend it uh and also a paper by inesh ipolito myself and our colleagues on Marco blankets in the braid so this is going to be a much too short summary of those discussions but basically you can start with individual

neurons and then

um you know talk about uh their Markov blankets

um to bring it back to something that I was saying earlier but that I didn't emphasize while we were going through the slides the Markov blanket identifies dynamical dependencies not necessarily physical boundaries so it's a statistical boundary it's not

necessarily like the boundaries of a wall like a cell in the case of formalizing individual neurons with Markov blankets we would associate the postsynaptic conductance with the internal States the post-synaptic voltage with the active States the pre-synaptic conductance to external States and the pre-synaptic voltage to sensory States

you have some equations of motion that tell you uh how each of these quantities change and what they depend on and you'll notice that this has the uh the dependency structure of a Markov blanket where the internal states only depend on uh sensory active and internal States um the active states only depend on internal States and so on so it respects the dependency structures uh that we were discussing a bit earlier where again you know the sensory states can only depend on other blanket States and external States and so on so here we have a nice Dynamic Markov blanket that we carve around our neurons and then there are two directions that we can go we can

Ascend the scales uh towards uh you know canonical micro circuits and then uh you know cortical columns uh individual brain regions brain whole brain networks and so on and at each of these successive scales of self-organization you can similarly identify uh Markov blankets

uh if you want to uh uh discuss this in more detail uh you can go see uh the paper by inesh and myself and colleagues um and you can also go from the individual neuron down uh towards um gated channels towards uh synapses uh dendrites and the uh Soma so the body of the cell all of these things have successive Markov blankets and so all throughout the brain's uh organization at the different scales at which itself organizes you have this nice structure of nested Markov blankets of Markov blankets of Markov blankets

um so that's one clear immediate sense in which there are these blankets and you know you can start to think of these blankets as screens uh in the way that we uh just uh we're talking about a slightly more illustrative example of what we're talking about here is that the Markov blankets and predictive coding this is a figure pretty typical in the predictive coding literature what it tells you is that the brain has a series of uh levels or layers where superordinate layers are basically passing predictions about sensory input onto subordinate layers and subordinate layers are shuffling

prediction error back up to the superordinate layers this is uh probably like you know I'm almost embarrassed again to discuss this at the Active Inference Institute I'm sure everyone is quite familiar with these things already the the novelty that we are introducing here is to say well each of these layers is connected to each other via a Markov blanket

um and so that's I think like a key Insight is that each of these layers has internal States and communicates two other layers through what you can formalize as a Markov blanket so you can understand at any given scale of brain activity you can understand the different levels of message passing as successively nested screens

the cool thing is that what you know what each of these screens is doing in some sense is coarse graining the uh stuff in the screen below so in some sense each screen is kind of implicitly contained in the following one and what you end up having is a kind of nested structure of screens of screens of screens of screens which is uh I mean pretty cool we recover as you'll note in this figure the

um the inner screen hypothesis each of these uh you know this is probably a better way to visualize it you have to at the left the uh out uh the kind of external Markov blanket of the organism it is uh segregating or carving out a set of internal States and what we're saying is these internal states have a

layered level level structure a sparse structure that we can formalize using nested Markov blankets um and like I was saying each screen successively coarse grains uh the next screen and this gives some flesh to this notion of contextual Computing this idea that really what the brain is is an organ of context where each layer is providing context to the layer below and where each layer is attuning to increasingly higher order uh slower and more

physically widespread regularities

um

so uh one thing that we're proposing here uh which is a novel hypothesis that we intend to test empirically over the next few years is that this architecture has two special layers in some sense there are two uh you can think about it as right only layers so if you're thinking about the stack

um the absolute well the the per the sensory periphery or the sensory bottom it only sense in one sense can only write in One Direction it can only write into the system

um and the topmost level

um can only write downwards and this is a key architectural feature one thing to point out is that in the paper we Associated uh the topmost level to structures like the hippocampus we don't need to suggest that the hippocampus is uniquely the top uh of this hierarchy rather there is probably a whole bulk of structures that uh basically do do not

cannot any longer be coarse grained so you kind of have like an executive level pop and what's particular about this top is that it can only write down into the stack so it's its function really is to be about and monitor the other layers of the stack implementing a form of covert action

and which trickles down uh throughout the stack so

um now I saw you write some stuff uh would you like to that yeah so I think one of the things that's important to highlight is why some things can only write up and some things can only like write down

um and what this means for irreducible the irreducible market linkage which is the part where if uh where our experience happens uh because there's a counter that and and some future questions that are going to happen that relates to Super structures and then why don't we experience the consciousness of a superstructure so I think it might be important to dig into that a little bit absolutely

um thank you uh so yeah uh to summarize then uh there are basically two right only layers uh as you ascend the layers which are successively coarse graining each other uh there is an irreducible screen that can't further be coarse grained uh we believe that this corresponds to something like a naturalized homunculus and we're calling this perspective it with an intentional provocation

we're calling this perspective

neocartesian

um in the sense that uh yeah we we may

have in some sense uh identified the

processes that make it seem as if there

was something like a homunculus

obviously uh we're not or maybe not

obviously it's worth saying we are not

um falling into the homunculus fallacy

because we're not claiming that the

topmost layers perceive themselves

so the buck stops with the top layers uh

and the top layers can only perceive

themselves vicariously by acting through

other layers

um and you know we we associate this

action of layers of internal Markov

blankets on each other with attention in

particular and other neural modulatory

effects

um so I mean the overall picture is one

where Consciousness is intrinsically

related to overt and covert actions uh

to the overt uh acting and sensing

associated with the external Marco

blanket and with the covert actions that

are being deployed within the stack and

in particular that are issued by the

very top uh set of levels as I said this

is not necessarily a strict hierarchy it

could be a hierarchy with several

different tops but this integrated top

structure we believe uh could be

associated with a naturalized homunculus

I'm going to skip through uh the rest

and thank you for your attention

uh unless now one wants to add something

just one thing is I think to put it in

in simpler terms as well
the two layers that can only write can't
store memories because that's right
memories is not
um it's not just an encoding it's a
process of
um of retrieval and it requires
interactions between different levels
which is why the lowest levels are only
interacting with things which are
external and therefore can't necessarily
store and we can't like make direct
interactions with anything else within
the The Entity itself and the highest
level as well so there's no memories
being encoded at these two levels which
is why they're right only and not um
yeah I think that that helps a little
bit

thank you yeah I would agree
um so uh so thank you all for your
attention thank you in particular Dan
for the invitation it's always great to
be at uh the idol a double I and uh yeah
we uh looking forward to this discussion
I think Daniel said he wasn't going to
uh to do too much um
um animation so uh if people have are
you gonna do uh are you gonna do any
facilitation at all then
yeah

okay so uh thanks Maxwell and Mao for
the amazing talk
um there are so many interesting things
to dig on in this paper I don't know
even where to start but I do have one
question too actually in particular
so uh does this in-screen model imply a

kind of Proto and psychism or a more restricted bio psychism I mean do you think the type of nested holographic structure uh described by the model only exists in biological systems and regardless of the answer would you think that it generalizes to larger systems comprising these systems like populations and colonies and so on maybe I can take a quick stab at that um so we uh we think that um the free energy principle itself commits you to a kind of um I don't know what the right word is we're kind of we're writing a paper now uh we're struggling with the exact coinage but it's something it's not quite a pan psychism it's it's a pen representationalism or a pen tracking ISM that you'll allow me to absolutely disgusting uh neologism um basically what the fpp implies is that anything that has a boundary is it looks as if it's tracking what's beyond the boundary so at a very basic level the fep kind of licenses this talk about anything so anything whatsoever looks as if it's tracking uh what's across the boundary by virtue of existing um so you know at a very basic level if I see a rock outside I know that you know it's not a thousand degrees Celsius outside right like the the fact that like a rock has a specific structure entails something about its environment uh likewise I can see uh you Maria you are not wearing a winter parka and I can uh you know infer from that that it's

probably room temperature around you and not like you're not somewhere in Antarctica or wherever else so uh this kind of basic tracking relationship applies to everything what we've tried to do in this paper is to start you know starts spelling out more explicitly what the difference is between uh you know just rocks and rock stars in some sense like what what is the special kind of structure that you need to add to the Markov blanket and what you'll notice is that you get Consciousness by making this the overall structure more sparse so as as you make it as you you as you make more of the connectivity zero you get all of these weird things that pop up one of which is we think like this this Consciousness structure um malice thought about this prop maybe more than I have so I'd be curious to get her perspective well so I know that people are a little bit um wary of pan psychism so we usually add the term Proto in front of it uh and so the idea is that we have to deconceptualize what Consciousness means and oftentimes what we mean is something that resembles what humans do uh and we have to be a little bit if we take our formalism Consciousness can become emergent at the scales where humans are where but but it doesn't mean that it doesn't exist at lower scales where uh the inertia or the entropy generated by policies is um is lower so if you think of a rock um

its policy is pretty much
maintain bonds you know I mean like
there's nothing else the scale at which
some kind of thing
exists and changes is much different
than you deciding to have coffee but in
in effect your cells could be something
that we could consider exist closer to
something that Iraq does and even lower
uh just the proteins that make up your
system so

um if we understand that there is
nestedness

um and that the the conscious process
happens once you have this capacity to
to exchange information across layers
anything that has this stacking of
layers in a nested fashion is in effect
um

proto-conscious but

we have to be very very careful about
what we load into the the term conscious

I think is is mainly the question

I'm I'm sharing my screen here this is
from a paper on path integrals

particular kinds and strange particles

uh so it is a typology of different

kinds of Markov blankets uh and

different kinds of particles

um so something like a rock it still has

a Markov blanket but it's it's Markov

blanket is not partitioned into sensory

and active states there are no states

that have this property that the active

states have that they affect the

external world that are not affected by

it in return

a rock just has sensory States if I kick

a rock unless it's Radioactive It's not going to act on me it's just going to uh it only has the inference part of active inference it's only kind of changing his State as a response to the forces that are acting on it

um active particles are particles that have both sensory and active States and you might think for example of a cell uh a cell it has this kind of structure um particles like ourselves

um have additional sparseness in the Markov blanket in particular what we call strange particles

um do not the internal states of strange particles do not have direct access to the active states of strange particles and therefore they need to be inferred and this is what gets you planning at the end of the day a strange particle can only exist in uh by kind of inferring itself into existence

precisely because there is there are no direct connections between the active and the internal States so as you remove uh connectivity

from the overall structure right uh here from here to here you're adding sparseness from here to here you're adding sparseness as as you make the system more and more sparse you get more and more interesting Behavior to emerge uh from the system and you know you can think of the topmost uh structure that we were describing as an instance of like this broader architecture uh where the topmost layer doesn't have direct access to the active states of the

system and therefore it kind of has to go through the entire bulk of the system to infer itself into existence

so there's a few hands up uh I don't there's no um it's not on Google where you can see the people who share it oh okay so Ali I think you raised your hand first

uh yes thank you again for your crystal clear and really illuminating uh talk uh I I've written down three uh questions but maybe I'll begin with uh probably the most basic and uh naive one uh so uh in some of the earlier literature on active inference and the fvp uh there was this emphasis on distinguishing between generative process and generative model

but in this paper uh in figure one which you also shown in your slides uh we don't see any generative process in the diagram so where does generative process fit into this diagram or uh because the way I see it uh I mean it's basically somehow included in the generator model or uh I don't know what is the the distinction between these two concepts so um you'll have to forgive some of our the earlier literature uh for this there's some terminological inconsistency uh the only two constructs that are at play here are the generative model and the variational density that's encoded by internal States that's it um so the it is not the case that the FTP is about how the generative model models the generative process the a lot of this language comes from uh you know

previous Bayesian approaches and in particular the helmholtz machine in the helmholtz machine uh so helmholtz machines are some of the earlier predictive coding machines and then the helmholtz machine basically you have a forward pass that connects the sensory end to this leave the top of the predictive hierarchy and then a backwards pass connecting it uh the uh you know the the uh the like top of the model back to the sensory end and in in a home holds machine you would call the forward pass a recognition model or uh and the backwards passage generative model right the forwards pass is a recognition model because you're inputting some data and then basically the by the time the information trickles up to the top you have recognized what caused the data and the backwards pass is called a generative model for the the reasons that are sometimes invoked in the predictive coding literature uh where you know you're basically uh generating possible sensoria given a certain configuration of States uh so the we don't use the recognition and generative in that sense anymore um and you know I apologize in some previous Publications I I've been a bit sloppy in the way that I use my language and I'd use the term recognition density what's really at stake is uh and I'll share my screen uh this to uh yeah the help make the point uh that's what it says here so you have a you have your generative

model right your generative model is a joint probability density so what is a joint probability density uh well you have a bunch of different uh basically states that your your system uh is defined over

um and you have a probability density uh that the shape of which captures the way that uh states can change over time right so like if I change this way then you know I need to change in that way and that's what this shape encodes

um and uh what the Feb says is if my generative model has a certain sparseness right if it has Markov blankets then there's this additional probability density that I can Define uh it is parametrized by the internal states of the system

and it's called the variational density and this is like my probabilistic best guess about what's causing my external States

um at no point do you need to appeal to any other bits of like yeah so generative process when you're actually coding these models up in Matlab or in Python what we call the generative process is just the equations of motion that govern external States

so if you actually hand code one of these one day your generative process might include like Newtonian physics and whatever else it's sort of just like what is the process that actually animates the Dynamics of external States um in in some there in some very simple models like in the uh the ones used by

Miguel Aguilera and colleagues you're looking at

um linear dissipative systems and they don't have any Dynamics right like so basically the the system kind of just it dissipates it's this a linear

dissertative system is a damped in

Spring physically right so it just goes

bring and then it dissipates to the

fixed point so similarly uh you may or may not have Dynamics to the external

States and that's what gets harnessed in

the generative process uh yeah this

should be clear yeah sorry man no I mean

I think what the question that Ali is

asking also has some

important

um

correlates that often get asked to us

like what's the difference between

Consciousness or Dreaming or what's the

difference between a dead system that

decays or uh being uh psychotic

um and I think what Max was saying is

our system currently doesn't say

anything about your actual connection to

the real world other than a thing which

is successful at being a thing is

probably well uh joined relative to its

needs to the world and therefore

um your relation to reality is probably

um has probably got a better grasp if

you are functional for a longer longer

amount of time

this minus senescence

um so yeah does that kind of

yes

yes thank you thank you both uh it it

really makes a lot more sense now thank

you so much

well uh it's my pleasure and remember

like model just means joint probability

density here right so uh

there's a a

criminally underappreciated paper that

Carl wrote uh in entropy in 2012 called

the free energy principle for biological

systems uh which is where a lot of them

are called blanket story comes from and

in that paper Carl refers to the

generative model as just a generative

density so it's basically saying there's

a there's a underlying probability

density it describes the way that the

Dynamics are just are being generated

just generally uh and yeah and then you

have the uh if if that density has the

appropriate structure meaning if it has

like zeros in the right places uh then

you can Define this this other tracking

density uh that approximates the

generative

um model or density

and I also have a follow-up question

which I'll ask after that Samuel and

thales

okay uh thanks uh for the excellent talk

uh

uh I'm a professor at the State

University of Minnesota

uh I'm a philosopher cognitive

scientists psychologists

so

I worked mainly on reasoning but

by reading your paper what strikes me as

most promising

is
the possibility of solving the the hard
problem of Consciousness if if this goes
right and progressing means this model
but maybe an important part of that
depends on exactly what is meant by
Holograms and exactly
what this brings into a a theory as an
explanation right so
doing your presentation you were talking
something about that there's a
difference between
holography in the sense of bringing
light to something to make a projection
uh but you said that might be related as
well so I'm quite curious about these
how to use this this concept and how to
understand it uh in a more
Soul psychology sense
and also like
where it comes from because I'm not a
physicist so does it have does it depend
on string theory to to
to be correct
could you dive in a little more on this
concept first
now I think this is more a you question
Theory so basically
remember Maxwell really gave a good
explanation I kind of thought I had to
answer your question but I'm gonna give
a little bit so the boundary between any
two mutually separation systems A and B
is the holographic screen that encodes
like the eigenvalues of the interaction
that couples A and B so uh this is very
similar to how a hologram encodes
information about 3D objects onto a 2d

surface

um so a hologram stores information about the phase and the amplitude of Lights reflected from an object and this allows the Reconstruction of a 3D image from a 2d surface and so here similarly the holographic screen uh encodes information about A System's interaction with its environment which allows for the prediction and control of the systems Behavior so does that yeah but does it maybe allow for the creation of an image that's that's the sort of question so this is yeah this is where we're giving it some thought actually so the creation of an image is it's a complex idea because what's an image right but effectively

um

there is some work being done into information geometry and how that may potentially interact with matter uh in in order to um

to to translate to transmit information basically uh and so it's possible that the Reconstruction of information is done through Fields being changed such that the modalities you have through your senses such as sight um hearing touch or relative to the way that the information needs to be transmitted in terms of amplitude and um and velocity Etc which in effect gives you something like light um but that research is still relatively um

early so yes it is promising it is

leading in that direction it's probably
one of the most interesting formalism
because it touches on the connection
between

um you know Dynamics matter and how
information

allows for matter to be organized
but all those pieces have been properly
being put together it is heading there
though

um I would just add um the so the
original monstrosity of a paper had a
section on uh yeah explaining where
qualia come from based on the
statistical patterns of the input but as
Mao said like it was very early work and
I mean the core model was more or less
developed so we decided to like spin
that out as a separate thing and then
develop you know the the inner con the
inner screen model of Consciousness per
se

um you know to flush out a little bit or
to add some flesh to what now it was uh
saying which was already pretty meaty as
it is

um so we we have you heard of these
experiments where they they take a
little uh

they take a little pad with little pins
on it and they basically reconstruct
well they they they plug a camera on
your head and they put the this little
pad on your tongue and if they blindfold
you and they they basically have you
navigate and the um the camera generates
a kind of um like grayscale
you know version of the image but with

the little pins in terms of like how far they push in uh Samuel are you seeing the image that I'm that I'm discussing or so you have a little little pad it has little pins that can push uh at various depths and what what you can do is basically translate the uh image from the camera into a different uh array of like assorted pins and so yeah you put this on your tongue and if you blindfold yourself then use this camera for about 20 to 30 minutes the brain eventually recognizes that oh this is supposed to go to the visual area and then then you start to see through your tongue in some weird way like the the the the body actually recognizes the statistical frequencies of the uh like of the pins actually pushing onto your tongue and and this is experience not um as like a a physical sensation but actually visually like you start to experience an impoverished grayscale visual field just from the sensations that are being fed to your tongue which suggests that there's something deeply in you know deep about the statistical structures of the signals that are being processed and the way that the qualia show up in your experience um so uh yeah uh but this is still like super far off in that I wouldn't I wouldn't have the pretension to claim that we've cracked or call you come from just yet yeah yeah uh I get what you're saying now uh

different uh patterns can uh make
different quality that you don't need
like different Magics to to have
different quality but
yeah I was wondering how that related to
to the Holograms but okay it's something
in development as you as you're saying
maybe I should uh let someone go and
then I'll go back to these Consciousness
questions
sure so Alice
well thank you all it was a great
presentation and great test by the way
hi Max thank you hi everyone I'm sorry
you guys are facing like a black screen
right now my camera is off that's all
right okay so
um I'm an anthropologists during
psychologist so I'm kind of an outsider
in here I'm not from the active interest
lab too so
you know I I study it but I'm not a
specialist in it so my questions are
kind of Broad and they're actually
really really simple but I think that
I'm not allowed to make one question
right pero and then
right so um actually my first question
will be uh something that Maria was
talking about before
me started so um the Articles account
Consciousness is an attribute of things
that possesses and reducible Market
blanket right with active States
exerting for instance over external
Dynamics so when applying to the neural
uh
uh physiological experience of the human

brain talking about neuromodulatory
influence over neuronal Dynamics uh and
with that in mind I would like to know
that should actually speculate a little
bit on the implication of this
structures issue of Consciousness for
how it might emerge during infant or
field development
that's a great set of questions uh we
gave this talk um
uh I gave a version of this talk to
um Luis pasoa's group at the
neuroscience and philosophy salon and
Anna Shanika asked a similar question I
would um I don't know I mean Mao
probably has a better response I would
point you in the direction of the work
of Anna shaunika
um she has some really cool work on
co-embodiment and cohomeostasis and
pregnancy and this kind of thing
um I uh I confess that I need to think
about uh the way of it like because
presumably you could just continue
extending this outwards right so we've
talked about inner screens
and there's like an external screen but
presumably you could just reapply this
whole screen talk in the other direction
like on the other side uh yeah exactly
the first prior well the the first prior
is um one of those papers but she's got
like another a set of papers uh yeah it
might this might be it right um so one
thing to also know could consider is the
sparseness aspect of um of our model
so
um as you develop you prune connections

and you become more and more sparse
which gives you more
um
path approximations
to the way that information gets
agglomerated to form patterns that are
recognizable
and so perhaps this has something to do
with the emergence of a moment where
um you sort of come online because
there's something like an irreducible
markup blanket that emerges by virtue of
having pruned enough such that you can
now make sense of the stimuli
relative to some couplings in time
rather than just hungry you know like
there might be something
um to that effect however
um
we need to be careful again with infants
what we mean about Consciousness like do
you mean
um
selfhood do you mean the capacity to
reflect on self put or do you mean the
capacity to
um learn and act on the world which is
uh closer to where we're headed because
um while babies may not necessarily have
this capacity to reflect on their
selfhood right away they do have the
capacity to react to signals and act on
the signals in order to get to a point
where their preferred states are enacted
so babies hungry cries food comes so
there is a degree to which a baby much
like a cell has a degree of
Consciousness relative to our model

so I'd also point you in the direction
of uh Rosalind Moran's work on the Aging
brain

um she's got some cool work that
suggests that you're well the Precision
of our model increases over time uh
which has to do a little bit with you
know what what Mal was discussing
earlier

um we we know just from neuroanatomy
that uh the process of pruning brain
connections uh only accelerates uh
through time and basically that this
that means that you're disconnecting
certain things from certain others right
like expertise has more to do with
understanding what is what is not
connected to uh than understanding what
is connected to what

um and so uh yeah there is probably uh a
kind of Developmental course whereby the
model becomes sparser and sparser and
sparser and sparser and you know more
and more precise uh but uh you know this
is a very cool direction for exploration
uh I I don't know if I have anything
more cogent to say at this time about it
pleasure

so I know Ali wanted a second girl and
uh so did Samuel does anybody else have
other questions because no other hands
are raised at this time

oh okay so we have one more hand I think
uh

I know uh actually uh you basically
answered uh my other questions uh so
thank you so much

pleasure awesome so then uh Dallas has

his hand up again Maria says her hand up
and seminal has hand up so tell us first
I suppose
I don't know you guys like to I know
it's Maria first because they just you
know
and then sure go ahead Maria
all right so
um yeah I didn't say before but I am a
psychologist and master in philosophy
and now I'm doing PhD in philosophy as
well and it's not really a question I I
would just like to to understand better
what
what is
memory in your in your inner mother
because uh I understand the the
information passing and the the middle
uh layers or levels that contains
information but uh how exactly do we
construct for for example The Narrative
self or something that we remember uh
from a long time or episodic memory I
mean I didn't understand it very well in
in the paper so if you could just shout
something it's a uh it's a it's a
computer science processing uh example
information processing notion of memory
um so uh you get some data and you need
to parse it
right into things that are causing that
data well presumably you need to
leverage some beliefs or you know
knowledge about how to parse that data
and that that comes in the form of
information basically uh yeah uh you
need to uh you know have some capacity
to reduce the entropy of your

observations and to kind of carve things out into things and the inner screen serves as the locus where the information is encoded so that's the cool thing about uh the quantum version of the uh markov blanket formulation it tells you uh where all of the information that characterizes the coupling across you know two bulks it where it lives basically I mean to get a little bit more technical the the Markov blanket itself is constituted by the degrees of freedom the couple the two systems uh together

um and yeah if you uh you can think of this as an array of bits that answer yes no questions about are the are the two systems communicating across this channel uh yes or no and in some sense the state of the blanket at any given time uh it gives you all of the information that you need to characterize the coupling uh in terms of these yes no questions so all of the classical information that you need to characterize the coupling lives on the blanket and

um yeah having an internal blanket means that you can have an internal store of such answers to yes no questions that you can then use to parse the sensory stream uh but in a in a kind of standard computer architecture it's just saying that you have a memory register right that you know so you yeah you need you need to store some of the information that you're processing somewhere so that you can deploy it and the idea is that

it lives on this internal screen
does that help and in this case it would
be the hippocampus
uh yeah and all well all of the inner
screens in code information in this
sense right the the point of having an
inner screen structure at all is to kind
of retain information to parse your uh
your sensory stream and yeah structures
like the hippocampus like these upper uh
brain stem structures or what used to be
called the limbic brain and so on we
think that these are all at the top the
idea right is that like so you have your
sensory motor end then you have your
prefrontal cortex which is the
controller for the pre for the sensory
motor and it's a controller in the sense
that its main function is to inhibit
activity in the sensory motor area but
um if if uh if the main function of
prefrontal Cortex is to go to the rest
of the brain then there is also a part
of the brain whose main function it is
to go to the prefrontal cortex and that
is these are these upper regions that
we're describing
um so yeah I the memory register is like
live in this whole stack
ultimately uh like the point about
memory is less unique to this Apex
structure what is unique to these top
structures though is the re the right
only kind of directionality that they
that they don't have any context right
like there there is no higher structure
that can provide memory to these uh top
layers so all they can do is act right

and and in particular they act in a kind of covert Fashion on other layers of uh the brain uh I think would be the answer to that now did you want to add anything to that

there's a

so you touched on the notion of selfhood and how the narrative cell unfolds over time and leverages memory and our model basically established it with self-hood the process of selfing is because according to our model and very uh like sorry I'm sick

uh very like

um Thomas metzinger's model of Southwood it's a process not a thing so selfing is the process by which the internal states are capable of mapping to the relevant external states such that they can act in such a way as to fulfill their own needs this is also very close to what Neil Seth about says about the beast machine

um and so if you have nested structure in each of these nested structure has a time scale say so you have structures that are that happen on the milliseconds and sometimes scales that happen on um this the scale of a sentence and then some structures that track much longer time scales like your life goals and uh and in your ancestry and stuff like that

um you find that

memory is just the story The binning of the relevant information at the time scale at which it is relevant and so you can retrieve the relevant information for the shorter time scales because it

acts as a prior for that for the lower
time scale does that help for the
narrative so yes all right thank you
awesome

um

sir go ahead

yeah

about the uh going back to the band
psychism problem uh in how you described
consciousness

I think it's

very well

uh

I mean it's old in the literature that
you can have information processing
which is not

uh

conscious but it's mental in some sense
in other animals or other or the beings
which are not biological beings
and but when you describe your models I
know you have this level also but you
also have a level of phenomenal
awareness or phenomenal phenomenal
Consciousness which you you are
describing

and

I think that's the most
particularly important part of the the
model the phenomenal awareness part
and

it's the model is the part of the model
that can contribute to something new to
me in in a sense that wasn't described
so I'm glad you mentioned matzinger
because Metzger is one of the few who
have described Consciousness as a
virtual reality

um

and what I think is interesting about describing Consciousness as a virtual reality is that it makes a difference between mental information processing or unconscious information processes in those processes that are conscious because they are simulated somehow uh

so an interesting thing of having a different language to speak of conscious events is that we can then explain why they are different and we can explain

why we have images in our head and stuff like that and why these images seem to have causal

Powers if they have at all so when I imagine something you can make my hand move or something like that and

you can describe this only with electrical information or something like that but

it doesn't explain why we have different information processes with

Consciousness and those without Consciousness and I know all Theory Of Consciousness have a distinction between the unconscious and the conscious level but adding something like the metaphysics or virtual reality would explain the metaphysics of conversation for instance if you're playing a video game

does the
the content of that game
matter
fertilization or is it just the
information that stored in the computer
that matters
I mean there's a a
variable there which means a sword does
it meaning a sword make any difference
at all
and I think that's something that hasn't
been very much discussed in philosophy
so David Chalmers has just written a
book
recently about virtual reality
but it doesn't it doesn't apply to to
Consciousness yet and
I think if you guys find something
uh in physics that's different
as as you guys are trying to do is you
admitted having that aim then maybe
there could be some sort of
a different physical explanation to
conscious causation so I would like you
guys to comment on that that's really
interesting we um we draw on medzinger's
uh work on opacity and transparency
um so you know in medzinger's work just
for the audience uh there are two kinds
of cognitive processes there are
transparent processes uh that we are not
aware of as cognitive processes per se
um and then there are opaque processes
uh which we are so for instance uh the
the metaphor comes from uh you know
trans a transparent or opaque window uh
and most of the time the window just
lets you see through it towards uh

whatever is on the other side but
sometimes it's a bit grimy and then you
see the window itself and so uh you know
similarly there are some processes that
mainly deliver things in the world and
then there are other processes that
deliver mental processes to us uh as as
kind of the object of our perception uh
in some sense uh so these um these are
processes like attention uh or like
paying uh attentional like processing or
paying attention to your own attentional
State uh these uh processes uh make
other processes opaque
um Terence Deacon uh the Anthropologist
also talks about similar phenomena in
terms of like absence absence and what
he calls absential phenomena so there's
a sense in which this top layer
what it does is Implement uh
opacification
so the the top layer can only be about
other layers in the stack and uh this
kind of reversal of the flow of
information is that we think like the
the key to understanding the way that uh
like mental processes are made available
for introspection at all
um yeah so uh that's that's how I would
respond uh Mao I saw you turn your
camera on uh do you have morta no no no
I don't have more to answer that
question
cool
um so
tell us you have another question
yep
so um I know guys ever really discussed

some of this before special on Twitter
um but it can just like repeat it
um because if I understood it right when
applying the model to human neurobiology
the article kind of sections
Central neural system
could you repeat the last thing you
broke you broke up pretty bad sorry my
laptop is really really interesting so
okay um the article seems to align
itself to a more modular account of the
central nervous system
um but in contrast to motor accounts
that some you know dynamicon distributed
approaches that suggests that there are
no parts of the brain with the Decay
functions right
um but regions with various capacities
that have expressed in software
assembled test specific collisions so
this correlation sometimes are termed
like transiently assembled local
neuronal subsystems right like yeah
we're big fans of like big families yeah
exactly so uh this approaches the
specialism is only a functional
Footprints and brain functions and I've
heard the result if you should fix it
networks implementing specific processes
but the result of high order relations
between regions so my question bear in
mind knowing that you guys know this
kind of approaches right
um do I think that the level VR
construction viable model could be cash
out in terms of transient correlations
built on the flight such as following
towns yes I mean we have a paper doing

that uh from 2020 it's called neural and phenotypic representations uh under the free energy principle and it is uh one of our lesser known um

uh papers let me just put it in the chat quickly

um and so what we show in this paper is that it's possible to simulate the form the transient formation of Markov blankets

um that basically track stimulus features uh so uh there are a lot of moving parts to what you were saying on the one hand yes uh you can totally use the Markov blanket formalism to model the transient formation and assembly of these

um I would also add that there's this there's a new uh perspective on modularity that opens up via the free energy principle

um the it's it's best explored in this paper by Thomas par inesh ippolito and a few others called modules or mean Fields where uh the argument uh and again this dovetails nicely with what Mike Anderson and his group are doing currently

um the argument is that uh the specialization of brain regions is not functional uh it's computational and this is effectively what Mike Anderson and Company are arguing is that like each each patch of the brain is specialized in converting certain patterns into other patterns it's not like you have you know a visual region or uh you know I think Mike's example is

the Broca's area which is associated with the production of language is actually activated in more non-linguistic tasks than in linguistic tasks so what that suggests is that each brain area is kind of like contributing certain stereotyped forms of computation to the overall process that's uh unfolding in the brain

um and in the 300 energy principle literature you can understand that as a kind of factorization right so what and where we have two different streams in the brain the dorsal and the ventral uh which respect which are handled the what I am seeing and where I am seeing and basically the the idea is that this uh anatomical segregation of Pathways corresponds to a factorization and inference where you are and what you are are independent you could be here you could be here uh and there yeah so uh I think I've tried to yeah

so that was a really good answer I'm going to try and say it a little bit differently and see uh see what sticks to the wall

um so

each hierarchical level of the brain has its like has Mark of blankets right so it's remember these things are nested um and they exchange with other levels according to the dependency structures of the mark of blankets

um and so this is like hierarchical predictive coding right so ascending prediction errors can be read as active States well descending can constitute

sensory States from the point of view of
a given level
um so using this factorization the
Maxwell just mention some brain regions
become specialized to encode distinct
probability factors that are
statistically independent and uh this
creates a sort of physiological
separation of brain regions which imply
that the the the information encoded in
these regions is conditionally
independent of that encoded in other
distinct regions uh so it allows for the
formation of new functions as the brain
regions interact
uh so we get to the point of mental
action this is enabled by the existence
of a subsystem
that is both
um
irreducible so there exists no further
Mark of blanket partitions of the
internal States and complex or
expressive enough to act like a suitably
flexible uh matter controller
so this matter controller coordinates
overt and covert action that's the last
part of what Maxwell explained which can
be seen as trans as forming transient
connections right so it that's the point
of precision uh Wayne
so the next the nested scales of
increasingly large ensembles of neurons
basically form
um coherently and so they also have
their own Mark of blankets
so we can generalize to different nested
connectivity structures in the brain

where we find uh
formally identical patterns at
subordinate levels and superordinate
skills
so does that
help restructure what Maxwell said
basically
yes yes it did awesome awesome
um so
several still has his hand up I think
Ali also has his hand up
um Ali didn't go yet Samuel do you mind
if Ali goes first
sure go ahead
uh thank you yeah I just wanted to
make a couple of comments about
the notion of self
self-encircling or self-modeling that's
just been discussed because
uh you see sometimes in philosophical
literature the sensibility of secondary
properties are usually uh they're
divided into the sensibility of things
in itself and Sensibility of things from
itself
and but under these conditions some
philosophers believe that real sensation
would not be possible or as Arthur Rimbaud
uh famously put it I
is an other right but in some recent
developments particularly in process
materialism
through the work of people like Thomas
Nagel the Manuel Delanda and Karen Barad
they've tried to somehow overcome this
issue this philosophical distinction
between in itself and For Itself by
introducing the notion of

self-receptivity uh name or in other words a capacity to be affected and to affect at the same time because uh by introducing a self-receptivity then the the sensation of each sensed point in the flow can become itself as an other to itself so basically that distinction between in itself and for itself would be dismantled effectively and that uh what that's why it can provide a more reliable way to discuss the notion of real sensation or real even subjectivity as a flow that just falls on itself and comprises both in itself and by itself and uh that's also why I believe

um fep and the active inference can provide uh even on the uh I mean more philosophical note uh more a reliable way to talk about what it means to uh to sense and be sensed at the same time I mean I I I find that very compelling I know you and Mao have been working together so I think Mao might have something more insightful than I I think that's great though

I agree nice

I agree

um

I think I like linking these ideas to what we wrote in the Consciousness text by basically saying that um

bigger screen is the folding onto itself right it's it can be seen as a as the wrapping inside another form the Consciousness is right into the form is wrapped in the form of the mind

um

and so you can if you partition internal
and external States and the market
blankets has these different kind of
active and uh sensory States

um

you can see how the folding basically
happens through this Loop uh that is
it isn't necessarily
happening because it's not only
happening at one scale does not have to
be non-simultaneous like we we consider
that uh that these things are
um separated at times simply because uh
we're thinking of one time scale
um but so if we if we add this notion of
interaction

um we can highlight the dynamic and
relational nature of Consciousness
um to refer to the process where the
agent and its environment actively shape
one another

um

so the Mind creating an inner screen
here just reflects its understanding of
the world and of itself through like
basically of the world through its own
filter and the filter is replica
reflection of itself it's just that
effectively if there are errors you need
to be able to model

what parts of

your filter

created potentially the error

um

so yeah

um you can yeah does that does that help
kind of flesh out this idea of holding

onto itself and incoming sensory data

I think this question was for Ali or

Maxwell

so one of you has to decide whether I I

thought it was for Ali that was great

okay cool

hahaha

yeah that was great thanks

awesome so Samuel I think you have a
question

yes

my final questions so

yeah I was mostly talking about the

heart problems courses because I I also

buy the predictive processing story and

I use what you guys are saying

uh but one issue that I have with the

productive process in the story is that

uh

we can apply it to explanations in

various ways so it's sometimes like

sometimes you can it seems you can tell
any story

with the same Theory and it's hard to

know which one is true if you can tell

multiple stories

so I think that's a problem with um

predictive processing in general and you

can explain for instance for instance

autism in different ways and I think

people have described that

um

your model

unconsciousness

how can we know that any of that is true

how empirical is it and during the talk

you guys mentioned something that you

were gonna test something

based on this which was very uh
promising and I'd like you to comment on
that so Maxwell has a really
he has a really cool paper on the map
territory fantasy fallacy which I think
is part of the answer here and we've
also written some uh so I'll let Maxwell
dig into it probably was going to be a
very flesh out answer so I'll give a
very small answer which he's going to
blow out of the water afterwards but
um

there are some things we can do to test
whether the predictions made by our
model actually come to fruition
um so that's that's one way to to test
the the empirical validity of a model
but to keep in mind a model logistic
model and so since you have multiple
realizability effectively what really
only matters is whether your model
actually has any predictive power
um so here let's say that we can
um examine whether the brain structure
and function reflects the nested and
hierarchical organization which is
predicted by our model
or whether intervention that alter the
brain structure or function lead to
changes in conscious experience as we
predicted
um or for instance we can test which
parts of the brain relate to which scale
of elements that are perceived and then
make predictions about the contents of
Consciousness and their neural
correlates to see how well it Maps so
these are things that we could

empirically test

I'll I'll just add a bit of meta flavor

to that

um so remember I started with the

distinction between Dynamics mechanics

and principles personally I don't like

the verbiage of predictive processing I

don't think it's a I don't think frankly

I don't think the term refers to

anything specific uh when people use the

term rigorously so

Andy Clark introduced the term in his

2013 paper whatever next because he

wanted to talk about some broad kind of

approach predict prediction involving

approach to explaining cognition but at

the time he he didn't want to commit to

any specific implementation of the idea

so uh he was like here's predictive

coding he was hierarchical predictive

coding but as I'm sure you know

predictive coding makes very specific

predictions about the neural

implementation of the theory so in

particular predictive coding assumes

that there are distinct neuronal

subpopulations that encode the error and

the prediction error he didn't

necessarily want to commit to one

version of that because who knows maybe

it's the same subpopulations that you

know issue predictions and prediction

errors at different times and you have

some kind of like time-based like rate

coding or whatever uh going on

um he also at the time didn't want to

commit to the free energy principle per

se so you know I think

yeah uh if the predictive processing doesn't really denote a specific Theory and one of the problems that emerged uh is you know there's a whole bunch of literature uh presenting accounts that are consistent with predictive processing which doesn't mean anything really uh it just means hey you can kind of wave your hands around and this kind of looks like a prediction if you kind of you know look at it the right way um I think the more rigorous way to understand this is to bring it back to the math and say that there are principles uh mechanics and Dynamics so there's some uh phenomenon that we want to explain we have a formal description of it this might be like a you know a data set or whatever else describing like the actual values of whatever system you're considering the free energy principle um is deployed to construct specific generative models uh which provide you with the equations of motion that explain that behavior right so the generative model is I mean in effect a set of linked equations of motions that you see here with these dependence relationships um and uh yeah so when you actually commit to a specific generative model then you are in a Terrain where you can bring empirical evidence to the to to kind of decide uh validate or to falsify the model so uh like I would draw your attention to you know the the free energy principle uh we know the free

energy principle is true for
mathematical reasons it would be as
silly to try to invalidate the fep
empirically as it would be to try to you
know falsify calculus by trying to
observe some phenomenon in the empirical
world it is true mathematically
and then you use that principle to
derive specific models which are then
testable uh so I don't I don't know if
we've gone all the way here uh so this
is somewhere still kind of in between an
in-principle explanation and a specific
model

um to get to more fleshed out empirical
predictions of the sort that Mao is
discussing we would have to get a bit
more specific which we're doing kind of
in parallel I I don't feel ready to
discuss the exact predictions and the
ways that we'll be um discussing them
this is uh or exploring them testing
them this is very much work in progress

I can tell you it'll involve
psychedelics

but uh that uh that's pretty much uh
where I'm coming I'm not comfortable
extending any more than that I feel like
I might expose a flank that's not ready
to be exposed

perfect thanks for that
pleasure

um and that brings us to the uh and uh
oh uh tell us you have your hand up
again uh do you wanna
hop on who has time
yeah yeah I think we'll have fun for one
more question yeah but Daniel you want

to ask something because whatever that's
two times
so okay um actually
I would like for guys to delve a little
bit deeper and the implications of
Maria's first question it's the other
side of the Quan
um related to the pen photo open cycle
stuff the population level conscious or
conscious processes in
complex organisms sharing a phenotype
for example like ant colonies or no more
speculative sense
um like
um complex organisms
um aligned and shared like a distributed
um generative model in Niche building
processes right so as Maxwell once said
the other side of the
um Metro selection
right process so
um given the sparsity and given the
transient you know
um conversions
that these populations are in this
specific moment specific process can we
speculate something about conscious in
this level
thank you
all right I think Maxwell is uh staying
quiet because he expects me to sort of
started answering the question correct
so um we've done a lot of work in
simulating large-scale systems right uh
so we've simulated uh Niche uh We've
simulated interaction between one and
two or more actors uh what it means for
belief sharing propagation the formation

of Market blankets Etc and so
um I think the key the true question
would be how transients are the market
blankets at the level of superstructures
and how does it enable those structures
to be able to encode
um
information at the level of the
interaction between different scales so
I do think that given our model you can
assume that a system is conscious by
virtue of having nested Parts which can
read and write where superordinates
entity or level of the the structure is
right only and the lower level is also
right only and through this um this sort
of boundary of the entity and the
interactions between the parts of the
entity you have what resembles a
conscious process at which you have some
degree of perception
um the real the really interesting
question there becomes what becomes the
irreducible parts so if you could be
understood a little bit like a cell if
you as a human your irreducibility is
that your cell as a part of a larger
system yeah
where what's experiences at the uh at
the system level uh and I think that's a
really truly interesting question if we
look at ants for instance
um
ants on their own don't do much
competition right but ends as a hive
then become much more interesting and
they behave uh in a coordinated fashion
which can look as if they were making

much more complex computation and you could say the same for yourselves your cells in themselves they have a very simple uh process very simple computation but

together through these large-scale emerging coordinating patterns some of them become your skin some of them become your brain uh and the in the the conscious part that you are experiencing at the scale at which you can understand Consciousness currently

um that's where most information most information is lost such that the irreducible part of your Consciousness is the parts that can become opaque um so

I think I've answered most of your question

now I would uh maybe just add the like so uh this is a uh not fully chartered territory and I think there may be some uh agreements and disagreements I am I'm always

um wary of the intentional misattribution fallacy uh where uh you know we attribute uh intentionality or mental properties to uh things that don't end up having them I think that the kind of minimal um

the kind of minimal intentionality pen intentionalism pen representationalism whatever you want to call it that follows from the free energy principle uh gets you uh a lot of the intelligent looking Behavior without having to pause it that there's actually something it's

like to be you know a city uh or
whatever what makes me skeptical of the
idea that there is such a thing is that
uh what what we are suggesting is that
what's necessary for Consciousness in
the stronger kind of agential sense is
an external Markov blanket that
insulates a bunch of internal States and
then these internal States in turn have
this kind of nested structure uh and
that's what I don't think that there is
in say like a city or uh like a higher
order kind of cultural I'd have to think
about this so this is you're kind of
pushing at the limits of where I'm
comfortable thinking which is
appreciated by the way thank you for the
challenge questions

yeah but I think there's room for
disagreement and I might be wrong about
this uh I don't think we just have to
keep pushing to find out who's correct
uh but I find what mala said uh
interesting and compelling and
intriguing I I am just like you know
intellectually conservative at this
point in my life being an old man
this is this is where I think it's a
little bit complicated to fully answer
that question because we we have
metaphorical tools but we don't have
anything ready yet to go any any any
further

um one thing that I like to think about
is

the only way you're ever going to
experience maybe what it's like to be
part of a Consciousness which is

entailed by

something more than you

um is having connection to sort of hyper

feed hive mind thing uh and being kind

of pushed and pulled so if you've ever

been part of like what a mob uh or if

you are on Twitter and you see people

um piling onto something and then you

talk to them and and they're like I

don't

I kind of blacked out I don't know what

happened because you you can't at this

at this part of the hierarchy compute

the entirety of what happened all you

can do is understand sort of your path

that was part of a larger network of web

of paths which themselves

um given the priors of the

superstructures you're embedded in are

kind of sort of going somewhere and

represent the

the error push of all the people that

share a similar type of model that

therefore constitutes a cultural

Direction uh so that's that's the best

way I think we can understand like hyper

priors as a function of what it might be

like to be a city sort of thing but

that's that's as far as I I can take

this metaphor for now

awesome

does anyone want to give a final thought

otherwise I'll just list just a few

points that I found really cool about

the discussion of kind of places it

takes us at the end but anyone else I

just want to say the the hard problem is

really hard and it's called like that

for a reason so uh like we were I think
we're you know hopefully moving in in
the direction of making this lesson
serious but uh like all of this should
be taken as very preliminary
investigations really trying to sketch
out the Bare Bones of an explanation
um and yeah there is far more work ahead
of us than uh behind us I think
um now any final thoughts
um I like to think about it in the
opposite terms
in terms of like the future of evolution
can be understood through like what it's
like to have language and what it's like
to have
um larger scale cognition and therefore
uh yes of course this is very
preliminary research but it's cool to
dream
awesome any guess Maria Samuel Ollie
feels
so any prediction on whether when the
other two proofreads will come out I'm
really interested in it
so the uh God you know inshallah the um
the mum paper will be released early
July
um right now reasonably
um and then the other one is has to do
with uh yeah extending outwards and that
might be like another year like I don't
think I don't think that's that's coming
soon
um but hopefully revisions to the this
uh paper will be coming out also over
the summer
um and yeah I would anticipate July

um

hopefully we'll we'll see there's a lot

going on right now so

awesome all right I'll just oh say no

please first

no I just want to say thank you to

everyone and for answering my questions

and yeah thanks

well thank you for your attention uh

your kind comments and your interest uh

it was really fun to discuss and I think

you asked a lot of important questions

bills

eventually or anyone else

uh yeah I

I also wanted to

uh thank you so much for having I mean

uh I know that these kinds of

discussions can have uh I mean lots of

implications in many many other areas

but it was really illuminating and

clarifying so

you're too kind my friend uh thank you

very much

um yeah this has been fun

genuinely

awesome all right I'll just give a few a

few comments

all right so first yes congratulations

to all for the recent papers it's really

exciting to see a kind of Continuum and

integration with all these different

threads

um the way that we think of physics and

physics of information

and the history and the philosophy of

the physics of information that kind of

a gradient might be reimagined such that

there's a clearer understanding of when
it's the firmware or the kernel of in
the tradition of as I see you at all
writing sometimes in versus
interpretations and and that's maybe
situational but that's a gray area
um reminds me of the instrumentalist
realism debates of 2021 your with the
map of the territory which has been
developed out so much further and just
the ability especially in a situation in
a model specific way to separate out the
intelligence or information structure or
the behavior from any kind of qualia
type discussion
um the recent
developments with uh Chris fields at all
and the waste energy and where free
energy goes and the kind of
saying like one has forgotten more than
someone else has learned so just the
idea that to have awareness there has to
be that discardle and then that might be
related to the irreversibility and the
time Consciousness that also you've been
uh developing
the eco-evo Devo components and just
multi-scale biology connecting it to
traditions of thought and work in
biology which is big but that's kind of
how we retro grandparent back into
biology promisingly with the ways that
multi-sale systems are already
considered and then also just the recent
developments in category Theory and the
Primacy of the relational and the
physics as information processing and
then the way that that enables very

knowledgeable individuals like you all
to come from a lot of the philosophy and
the history and all these other threads
to come together with the actual
modeling
in a very satisfying way so thank you
all for the stream and really looking
forward to the next steps
bye thank you bye-bye
thanks